Angle Pairs: Supplementary, Complementary, Vertical, Adjacent

Teacher's Edition

The Period

MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical or adjacent angles.

Must Name adjacent, vertical, complementary and supplementary pairs (a linear pair is supplementary); find a missing angle from each; write and solve the equation when the angles are expressions, and report the angle the question asks for.

Must not Parallel lines and transversals (Geometry); angle sums of polygons (MA.8.GR.1.6); reflex angles.

Target I can name an angle pair and use it to find a missing angle, with an equation when the angles are expressions.

MTR **MTR.2.1** — Notes II–III: one pair as a picture, a name and an equation · **MTR.4.1** — boards 4 and 7 split; the re-vote · **MTR.5.1** — Notes III and boards 5–6: the same equation shape behind every pair

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	10 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	12 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. Adjacent angles share a side; vertical angles sit across a crossing and are equal.
2. Complementary makes 90 — a corner; supplementary makes 180 — a straight line.
3. Name the pair, write the equation, solve — then put x back: the question asks for the angle.

Slide by Slide

1. **Angle Pairs: Supplementary, Complementary, Vertical, Adjacent.** Post the learning target.
2. **Warm-Up — On your own. No notes.** Q4 is today's 'supplementary' before it has a name: a straight angle is 180°, so the other is 50°. *1 retrieves 6.07 — a shadow; yesterday was the test, so the band reaches the last lesson (yesterday). 2 retrieves 6.05 — same y, across zero (last week). 3 retrieves 5.10–11 — a unit rate (last unit). 4 retrieves MA.4.GR.1.3 — angle measures add; a straight angle is 180° (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 27 ft 2. 9 units 3. 42 4. 50°
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the names and the picture.** Next to, sharing a side: adjacent. Across: vertical, and equal.
6. **Notes.** Complementary: corner, 90. Supplementary: straight, 180.
7. **Notes.** Name it, write it, solve it, put x back.
8. **Example 1 — A right angle, split.** Ask what the square mark says before anyone writes: the two angles make 90 — complementary.
9. **Example 1 — Worked.** Complementary: 3x − 1 + 2x + 6 = 90, so 5x + 5 = 90, x = 17.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 4x + x + 10 = 90, 5x = 80 and x = 16, so 64° and 26° — a board with 16 stopped at x; one with 34 used 180.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Vertical angles.** Ask what vertical angles do: they are equal — so the two expressions are set equal.
13. **Example 2 — Worked.** Vertical, so equal: 5x − 12 = 2x + 18; take 2x off both sides, 3x − 12 = 18, so 3x = 30 and x = 10.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 4x + 3 = x + 39, and taking x off both sides gives 3x + 3 = 39, so x = 12 and each angle is 51° — a board with 12 reported x.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **16–17. Board 1.** What is the complement of 34°? → ***56°*** *146° — used 180 [Florida B1G-M, GR.1.4: complementary and supplementary swapped]*
17. **18–19. Board 2.** What is the supplement of 112°? → ***68°*** *−22 or 'none' — used 90 [Florida B1G-M, GR.1.4: complementary and supplementary swapped]*
18. **20–21. Board 3.** Two lines cross. One angle is 47°. Find a and b. → ***a = 47°, b = 133°*** *a = 133°, b = 47° — vertical and adjacent swapped [Math Nation names thinking every adjacent pair has a special sum]*
19. **22–23. Board 4.** $\angle P$ and $\angle Q$ are complementary, and $m\angle P = 28^\circ$. What is $m\angle Q$? → ***B — 62°*** *(A) 180 − 28: supplementary for complementary [Florida B1G-M, GR.1.4: complementary and supplementary swapped]; (C) took the two as equal — that is vertical angles [Notes I]; (D) 90 + 28: added instead of subtracting [Notes II: the two add*

to 90] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

24–25. Board 5. A linear pair measures $(3x+15)^\circ$ and 105°. Find x. → ***x = 20*** *30 — set 3x + 15 = 105, as if vertical [Math Nation names setting a linear pair equal]; −10 — used 90 [Florida B1G-M, GR.1.4]*

26–27. Board 6. Vertical angles measure $(6x-20)^\circ$ and $(2x+28)^\circ$. Find the measure of each angle. → ***52°*** *12 — reported x as the angle [Math Nation names reporting x]; 21.5 — added them to 180 [vertical angles are equal, not supplementary]*

28–29. Board 7. Which two angles are adjacent? → ***C — angles AOB and BOC*** *(A) a vertical pair: across the crossing, no shared side [Notes I]; (B) angle AOB lies inside angle AOC — the two overlap [Notes I: no overlap]; (D) these share only the vertex O, not a side [Notes I: a shared side] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

30–31. Board 8. Above a straight line a ray makes angles of 38° and x°; below it another ray makes an angle of 50°. Find x. → ***x = 142*** *The 50° angle below the line is not needed. 92 — used the 50° (180 − 38 − 50) [the extra number]; 52 — used 90 [Florida B1G-M, GR.1.4]*

32–33. Board 9. Can two vertical angles be complementary? Always, sometimes or never? Explain. → ***Sometimes — only when each is 45°.*** *'Never — they are not next to each other' [complementary is about the sum, not about being adjacent]; 'always' [Florida B1G-M, GR.1.4]*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–2: a complement, then a supplement; 3 reads a crossing (vertical and a linear pair at once); 4 votes on 90 against 180. The pattern breaks at 5 — a linear pair with x — then vertical angles with x where the angle is asked (6), which pair is adjacent voted (7), and a picture with one angle too many (8); 9 is written. The independent set: a complement, a supplement, a crossing, a linear pair with x, vertical angles with x, a pair named and used.

Misconceptions to Watch

Board 1. 146° — used 180 [Florida B1G-M, GR.1.4: complementary and supplementary swapped]

Board 2. −22 or 'none' — used 90 [Florida B1G-M, GR.1.4: complementary and supplementary swapped]

Board 3. a = 133°, b = 47° — vertical and adjacent swapped [Math Nation names thinking every adjacent pair has a special sum]

Board 4. 180 − 28: supplementary for complementary [Florida B1G-M, GR.1.4: complementary and supplementary swapped]

Board 5. 30 — set 3x + 15 = 105, as if vertical [Math Nation names setting a linear pair equal]

Board 6. 12 — reported x as the angle [Math Nation names reporting x]

Board 7. a vertical pair: across the crossing, no shared side [Notes I]

Board 8. 92 — used the 50° (180 − 38 − 50) [the extra number]

Board 9. 'Never — they are not next to each other' [complementary is about the sum, not about being adjacent]

Materials Whiteboards and markers. Calculators allowed, printed nowhere.